

PROGRAMMING FUNDAMENTALS

Spring 2024

5th April

Variable

- A variable is a name given to memory location that a program can use

Declaring variables

- There are three places where you can declare variables in a program:
- Inside a function or a block: Local variables
- Outside of all functions: Global variables
- In the definition of function parameters: Formal parameters

Local variables

```
int main()  
{  
    int a;  
    int b;  
    float c;  
    ...  
}
```

- A local declaration is made inside the body of a function. Locally declared variables cannot be accessed outside the function they were declared in.
- The variables in main() are local, local to 'main' function
- It is possible to declare the same variables in other functions

Local Variable

- A local variable exists until the block/ the function is in under execution
- After that, it will be destroyed automatically
- Local variables are not known to functions outside their own

```
int add_func (int a, int b)
{
    int sum;
    sum = a + b;
    return sum;
}
```

Global Variables

- Variables defined outside the bodies of all functions and outside the `main()` program
- Placed at the beginning of the program file
- The global variable will hold its value throughout the lifetime of a program
- They can be accessed within any function defined for the program

Ex-1

```
#include <iostream>
using namespace std;
int func1(int x);
// Global variable declaration:
int ans;

int main () {
    // Local variable declaration:
    int a, b;
    a = 10;
    b = 20;
    ans = a + b;
    cout <<"Ans in main unction = " <<ans<<endl<<endl;
    int f_ans;
    f_ans = func1(a);
    cout<<"Answer coming from function1 = " <<f_ans;
    return 0;
}
int func1(int x)
{
    return x + ans;
}
```

Ex-1 (Solution)

 D:\Books-ICT\C-Programs\Lac14-PracticeCodes.exe

```
Ans in main unction = 30
```

```
Answer coming from function1 = 40
```

```
-----
```

```
Process exited after 0.1158 seconds with return value 0
```

```
Press any key to continue . . .
```


Ex-2

tled1 [*] Lec-16.cpp

```
#include <iostream>
using namespace std;
int func1 (int x);
int ans=0;           // global variable
int main()
{
    int a,b;
    a = 10;
    cout<<"The value in ans in the main function = "<<ans+a<<endl<<endl;
    int f_ans;
    cout<<"ans = "<<ans<<endl;
    f_ans= func1 (a);
    cout<<"The value in ans coming from the function - func1 = "<<f_ans;
    return 0;
}

int func1 (int x)
{
    return x + ans;
}
```

Ex-2 (Solution)



E:\My Course Books\ICT\C++File-F2023\2024\Lec-16.exe

```
The value in ans in the main function = 10
```

```
ans = 0
```

```
The value in ans coming from the function - func1 = 10
```

```
-----
```

Ex-3

```
#include <iostream>
using namespace std;
    int func1();
// Global variable declaration:
int ans;

int main () {
    // Local variable declaration:
    int a, b;
    a = 10;
    ans = a*a;
    ans = ans + 10;
    cout <<"Ans in main unction = " <<ans<<endl<<endl;
    int f_ans;
    f_ans = func1();
    cout<<"Answer coming from function1 = " <<f_ans;
    return 0;
}
int func1()
{
    return ans + 10;
}
```

Ex-3 (Solution)

 D:\Books-ICT\C-Programs\Lac14-PracticeCodes.exe

```
Ans in main unction = 110
```

```
Answer coming from function1 = 120
```

```
-----
```

```
Process exited after 0.1265 seconds with return value 0
```

```
Press any key to continue . . .
```

Ex-4

```
#include <iostream>
using namespace std;
    int func1();
// Global variable declaration:
int ans;

int main () {
    // Local variable declaration:
    int a, b;
    a = 10;
    ans = a*a;
    ans = ans + 10;
    cout <<"Ans in main unction = " <<ans<<endl<<endl;
    int f_ans;
    f_ans = func1();
    cout<<"Answer coming from function1 = " <<f_ans;
    return 0;
}
int func1()
{
    ans = 1;
    return ans + 10;
}
```

Ex-4 (Solution)



D:\Books-ICT\C-Programs\Lac14-PracticeCodes.exe

```
Ans in main unction = 110
```

```
Answer coming from function1 = 11
```

```
-----
```

```
Process exited after 0.1296 seconds with return value 0
```

```
Press any key to continue . . .
```

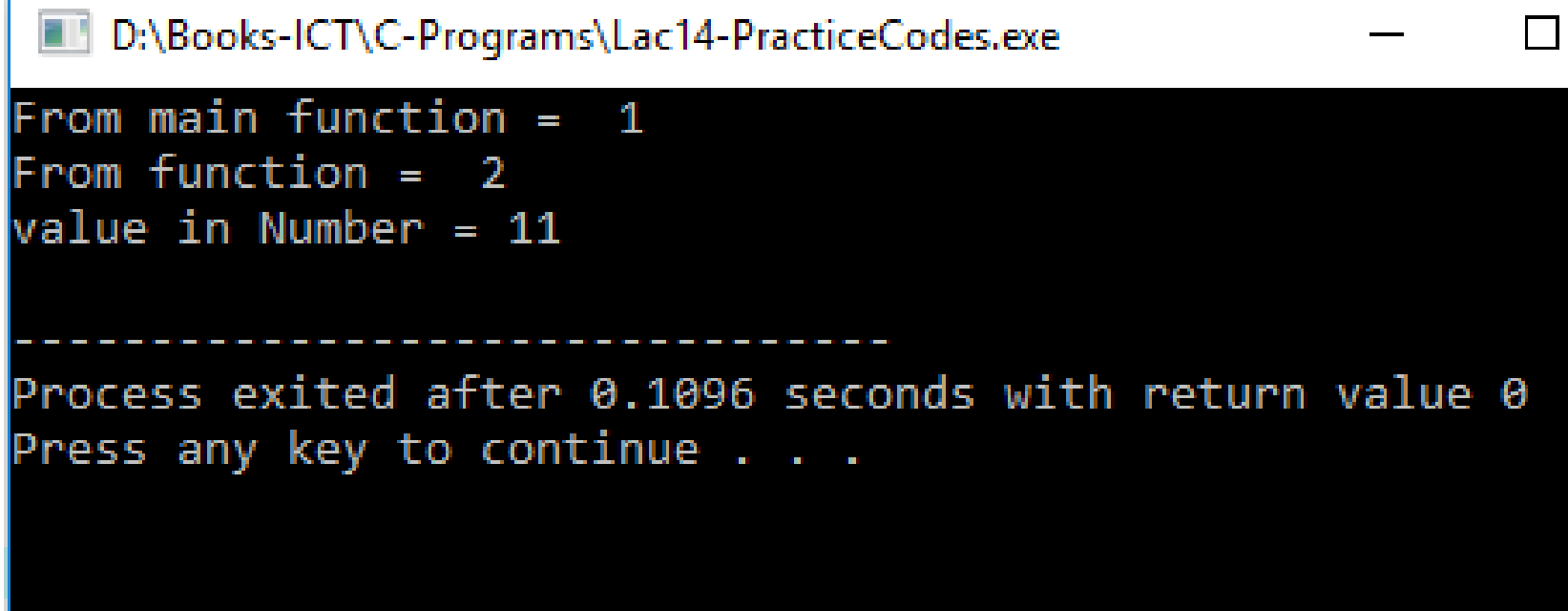
Ex-5

```
#include <iostream>
using namespace std;
int Number; //global variable

void my_func(int Num)
{
    Num = Num + 1;
    cout << "From function = " << Num << endl;
    Number = Number + 10;
    cout<<"value in Number = " << Number<<endl;
}

int main()
{
    Number = 1;
    cout<<"From main function = " << Number<<endl;
    my_func(Number);
    return 0;
}
```

Ex-5 (solution)



```
D:\Books-ICT\C-Programs\Lac14-PracticeCodes.exe
From main function = 1
From function = 2
value in Number = 11

-----
Process exited after 0.1096 seconds with return value 0
Press any key to continue . . .
```


Ex-6

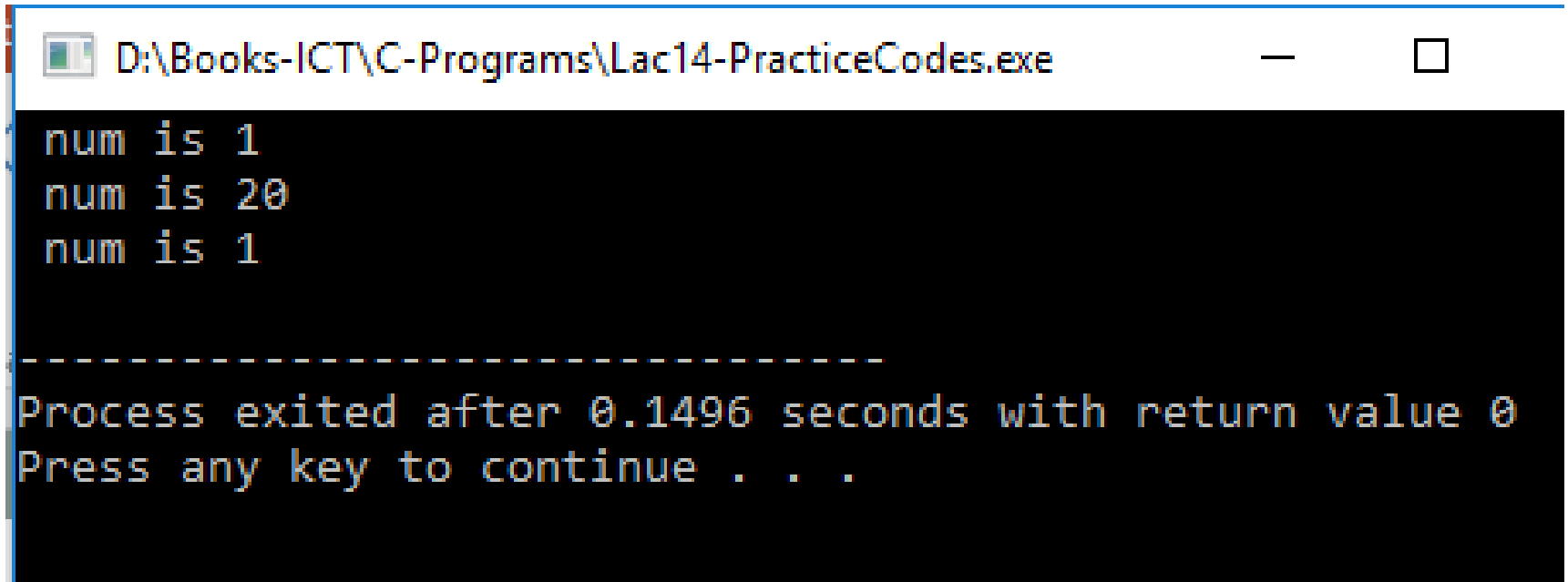
```
#include <iostream>
using namespace std;

void anotherFunction(); // Function prototype
int main()
{
    int num = 1;

    cout << " num is " << num << endl;
    anotherFunction();
    cout << " num is " << num << endl;
    return 0;
}

void anotherFunction()
{
    int num = 20; // Local variable
    cout << " num is " << num << endl;
}
```

Ex-6 (Solution)



A screenshot of a Windows command prompt window. The title bar at the top shows the file path "D:\Books-ICT\C-Programs\Lac14-PracticeCodes.exe" and standard window controls (minimize, maximize, close). The command prompt has a black background with white text. It displays the output of a C program: "num is 1", "num is 20", and "num is 1" on three separate lines. Below these lines is a dashed line separator. At the bottom, it says "Process exited after 0.1496 seconds with return value 0" and "Press any key to continue . . .".

```
D:\Books-ICT\C-Programs\Lac14-PracticeCodes.exe

num is 1
num is 20
num is 1

-----
Process exited after 0.1496 seconds with return value 0
Press any key to continue . . .
```

Local Variable Lifetime

- A function's local variables exist only while the function is executing
- This is called the *lifetime* of a local variable
- When the function begins, its local variables and its parameter variables are created in memory
- When the function ends, the local variables and parameter variables are destroyed
- So values stored in a local variable is lost

Global Variable

- A global variable can be accessed by all functions that are defined after the global variable is defined

Parameter	Local	Global
Scope	It is declared inside a function.	It is declared outside the function.
Value	If it is not initialized, a garbage value is stored	If it is not initialized zero is stored as default.
Lifetime	It is created when the function starts execution and lost when the functions terminate.	It is created before the program's global execution starts and lost when the program terminates.
Data sharing	Data sharing is not possible as data of the local variable can be accessed by only one function.	Data sharing is possible as multiple functions can access the same global variable.
Parameters	Parameters passing is required for local variables to access the value in other function	Parameters passing is not necessary for a global variable as it is visible throughout the program
Modification of variable value	When the value of the local variable is modified in one function, the changes are not visible in another function.	When the value of the global variable is modified in one function changes are visible in the rest of the program.
Accessed by	Local variables can be accessed with the help of statements, inside a function in which they are declared.	You can access global variables by any statement in the program.
Memory storage	It is stored on the stack unless specified.	It is stored on a fixed location decided by the compiler.

<https://www.guru99.com/local-vs-global-variable.html>

Book Reference

- Tony Gaddis - Chp-6
- D.S Malik - Chp - 6